

Assignment1 ESDM

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SECP3744: ENTERPRISE SYSTEM DESIGN AND MODELING (WBL)

Technical Report

Topic: How AI-powered analytics improve forecasting, resource allocation, and customer service

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Executive Summary

The report discusses the AI-powered analytics and how it can transform the antiquated enterprise platforms to become predictive and prescriptive intelligence engines, which responds to the demand of improving forecasting, resource distribution, and consumer support. It is revealed that AI gives a specific competitive edge, as it is evidenced by the examples of Amazon, Walmart, and Walgreens. Through machine learning, external data and real-time simulation, AI can be used to perform accurate forecasting, optimizing resources dynamically, and offering customer services in a hyper-personalized way. Even though the advantages in terms of efficiency and accuracy are evident, there are serious obstacles, such as the high cost of its implementation, the acute need to rely on unbiased data, the problem of the black box that introduces ethical risks, and the complicated integration of the legacy systems. It has been concluded in the paper that the implementation of AI is a strategic requirement and proposes a gradual approach to implementation, focusing on data governance, and investing in Explainable AI (XAI) and general workforce reskilling to create a clear and human-friendly Intelligent Enterprise.

1.0 Introduction

In the modern digital economy, businesses are overwhelmed by enormous amounts of data produced by all areas of their activity. Enterprise systems such as Customer Relationship Management (CRM) and Enterprise Resource Planning (ERP) are the nerve centres of the new organisation, and the amount of data accessed by them is often too much and too difficult to be processed by other traditional analytical systems. It is difficult to overestimate the importance of effectively utilising such data, as it forms the foundation of competitive advantage, operational efficiency, and informed business strategies. In this aspect, the artificial intelligence (AI)-based analytics has become a disruptive technology, which enables businesses to leave behind the history of reporting and move to the world of predictive and prescriptive intelligence.

Traditionally business intelligence has been rooted on descriptive analytics which presents a rear view image of performance. The conventional forecasting techniques such as the use of the past sales history and the heuristic forecasting models have been identified to be poor in capturing the dynamic consumer demand [1]. The methods cannot explain all important factors, including seasonality and promotion, thereby leading to forecasts errors. This may lead to expensive overstocking or harmful stockout, which is harmful to the financial health, as well as customer satisfaction. The fact that these legacy systems cannot adapt rapidly to market changes underscores the fact that more sophisticated analytical features are in demand, thus limiting the capabilities of older models becomes even more evident in a dynamic environment [2].

The main focus of this paper is to provide an in-depth research on how AI-based analytics are transforming the core roles of modern business organisations. Its scope is centered on three key areas, namely to improve accuracy of demand forecasting, resource allocation efficiency,

and efficiency and personalisation of customer service. This paper will attempt to give a clear understanding of the strategic importance and practical importance of these technologies within the contemporary enterprise setting by exploring their mechanisms, application as well as implications.

2.0 Background/ Literature Review

Active, intelligent decision-making has replaced passive data collecting in enterprise analytics. In the past, traditional BI offered diagnostic and descriptive insights from static data, answering “what” and “why”. This reactive approach limited enterprises.

By offering predictive and prescriptive capabilities, AI-driven analytics has transformed this, allowing organisations to expand past historical comprehension to foresight. By recommending and even automating data-driven activities, these complex analytics transform the enterprise system from “System of Record” to “System of Intelligence”, allowing for better and more flexible decision-making.

Another significant driver is “Embedded AI”, which integrates machine learning models into products and processes. Large language models (LLMs) improve access and eliminate technological barriers by simplify data interaction. Employees and business leaders can now use natural language to extract insights, promoting a data-driven culture.

The “Intelligent Enterprise”, where smooth technology and data integration produce a flexible, responsive and proactive organisation, is created by the convergence of AI, IoT and core business data.

3.0 Analysis and Discussion

3.1 How the technology works

AI enhances forecasting by implementing advanced machine learning models (ML). Unlike traditional methods, which are limited to internal historical data from an ERP system, AI models operate on a broader scale. The AI’s main function is to analyse past data in combination with current external sources. These can include economic data, social media sentiment and weather forecasts. A systematic review by Antwi (2019) supported this, concluding that AI produce much more accurate results than conventional forecasting techniques because of its capacity to integrate and learn from complex data.[12] By identifying the complex relationship between all of these internal and external variables, the AI produces a forecast that is more dynamic and accurate.

By moving beyond rigid and worksheet-based planning, AI improves resource allocation . AI create schedules using simulation and optimisation algorithms. These systems analyse large

amounts of real-time data such as current inventory from ERP/SCM systems, and equipment availability from IoT sensors. The best resource distribution, such as which equipment to use, which materials to select, and which operators to assign is determined by the AI then runs thousands of "what-if" scenarios. This ultimately saving costs, reducing downtime, and meeting deadlines. This capability is particularly valuable in energy-intensive sectors. Furthermore, financial management is now a part of AI-powered resource optimisation has expanded into it.[10]

AI enhances customer service within enterprise systems by employing sentiment analysis and natural language processing (NLP). First, all incoming customer communication is automatically analysed using sentiment analysis. The system can recognise and classify the customer's emotional tone, such as "positive", "negative" or "frustrated". This capability allows more intelligent routing, like a query classified as "angry" can be automatically forwarded to a senior human agent, while AI-powered chatbots or virtual assistants will 24/7 handle simple and repetitive questions. Lastly, AI increases agent productivity. The system can act as an agent assistant that provides suggested responses in real time. After an interaction, it can automatically summarise the conversation, reduce manual work and improve record accuracy.[11]

3.2 Case Example and Applications

An excellent illustration of AI-powered forecasting is provided by **Amazon**. The business uses a new foundation AI-forecasting model to forecast customer demand across millions of items, specifically what products consumers will want, where they will take them and when. Compared to Amazon's previous system, which mainly relied on past sales data for inventory planning, this new strategy is a major improvement. The main benefit of the AI model is its capacity to integrate and analyse time-sensitive external data, like weather patterns and holiday schedules, to optimise product arrangement. Significant outcomes from this AI application include a 20% increase in regional estimates for popular items. As a result, the company's carbon footprint is decreased, shipments can arrive faster and delivery partners must travel fewer miles.[14]

Walmart use Artificial Intelligence to optimise its huge supply chain and operational procedures as an example of resource allocation. The company deploy AI-powered robots in its warehouse to automate important physical tasks like product sorting and packing. Together with human employees, these robots improve overall accuracy and efficiency. The use of AI for automated inventory management and supply chain optimisation successfully streamlines operations and reduces the time and resources required to manage inventory and fulfil customer orders.[13]

The case example for customer service is **Walgreens**. They focus on their AI transformation on creating "comprehensive customer profiles" to provide mass personalisation. This is achieved using advanced agent-assist skills for its pharmacists and 24/7 self-service options,

including chatbots driven by AI to answer general queries. As an obvious example, the AI system can notify a client through their app when the prescription is ready and, at the same time, show a prompt on the pharmacist's screen indicating that the customer's family member also needs to get a flu shot. This demonstrates a clear change from reactive to proactive consumer involvement by enabling the pharmacist to offer proactive, relevant and highly personalised healthcare service during the conversation.[15]

3.3 Advantages and Disadvantages

The primary benefit of AI analytics is a massive boost in the accuracy of decision-making and efficiency of operations.. Forecasting enables precise demand prediction by analysing large datasets, which optimises inventory and reduces costs. AI makes it feasible to allot resources dynamically based on real-time data to optimally increase productivity. Furthermore, AI in customer service offers 24/7 customer support through automated tools and facilitates massive personalisation through data insights that boost customer loyalty [3]. These advantages constitute a basic competitive advantage of the enterprise.

Despite the enormous advantages, there are several disadvantages to implementing AI. High upfront costs in infrastructure, technology, and specialised talent are a huge barrier [4]. Secondly, the quality of an AI algorithm relies entirely on the quality of training data; poor or biased data will lead to biased conclusions. Furthermore, some complex models can create issues with the lack of transparency in their decision-making processes, which is known as the "black box" problem, and creates trust and ethical risks [5]. Finally, incorporating AI into a company's legacy systems is usually a difficult and prolonged process.

3.4 Challenges and Future Trends

There are various critical challenges that may hinder the effective integration of advanced analytical systems. One of them is reliance on high-quality and complete data; systems trained on partial or biased data will always generate incorrect insights [6]. This is normally complicated by the fact that the technology itself is a heavy investment, as well as the specialised knowledge that is necessary to operate it. Moreover, the deployment of these novel platforms with the pre-existing legacy enterprise systems is technically and logistically challenging. Lastly, companies have to operate within a multifaceted environment of ethical considerations and legal demands, especially regarding data protection and the openness of automated decision-making procedures.

In the future, the development of analytics is aimed at increasing its autonomy, transparency, and human-machine interaction. One of the key trends is the major change in the concept of predictive analytics (anticipation of future events) to prescriptive analytics (suggesting the most appropriate course of action). In this paradigm, the systems will forecast demand, as well as automatically propose or even implement optimal strategies to allocate resources, in real-time [7].

The increasing focus on Ethical and Explainable AI (XAI) is to deal with the issue of the black box. This includes creating clear, auditable algorithms to promote fairness, reduce bias and create organisational trust [8]. Lastly, this technological change requires the restructuring of the workforce. It is no longer limited to a small number of experts but encourages the idea of more general AI literacy, whereby employees in various departments are reskilled to read and work with AI-based insights, toward a model of AI-enhanced decision-making.

4.0 Recommendations

Based on the analysis, several suggestions are proposed for companies looking to integrate AI-powered analytics. First, data integration and governance must be prioritised. The report identifies that the AI system quality depends just for high-quality, accurate data because biased data will cause the system to produce inaccurate conclusions. Therefore, organisations must invest in excellent data cleaning procedures and methods for integrating with older systems, which is frequently a challenging procedure. Second, it is advised to employ a phased implementation method. Organisations should begin with a single, high-impact area, like demand forecasting, to show a clear return on investment before expanding to other areas, given the considerable barrier to infrastructure and specialised expertise. Finally, companies need to invest in workforce reskilling and Explainable AI (XAI). To address the “black box” issue, which increases ethical and trust issues, leadership should give priority to the XAI trend to guarantee that algorithms are transparent and auditable. This technology change also required workforce restructuring, and employees must reskill to achieve general AI literacy to work with AI-based insight.

5.0 Conclusion

AI-powered analytics has transformed enterprise operations, shifting business from reactive historical analysis to proactive, intelligent decision-making. This report demonstrates how AI greatly improves the accuracy of demand forecast, optimises resource allocation and customises customer support. The success of businesses like Amazon, Walmart and Walgreens shows that companies can now predict market movements, use resources efficiently and provide a highly customised consumer experience due to advances in machine learning, external data integration and predictive skills.

Although AI has many benefits, like increasing customer loyalty, cost savings and operational efficiency, it remains a challenge to overcome in its application. High upfront expenses, the crucial requirement for high-quality data, “the black box” issue, and the challenges of integrating with older systems are some of the main challenges. However, these challenges can be addressed through prioritising accurate data governance, implementation strategy and investing in workforce reskilling and Explainable AI (XAI). To create an “Intelligent Enterprise” where AI literacy supports a data-driven culture and eventually results in sustained competitive advantage in a dynamic global market, the enterprise system of the future will employ AI that is more autonomous, transparent and human-centric.

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Logbook:

<https://github.com/lauyeewen123/Enterprise-Systems-Design-Modelling/blob/main/README.md>